Name: Shashwat
Class: API-232 Computational Policy Analysis: Data, Decisions, and Evidence

ASSIGNMENT 1 – Descriptive Data Analysis

1. P(E) = 5%
   P(F|E) = 80%
   P(F|Not Error) = 10%

   a. Probability of being flagged given the application contains an error = P(Flagged|Error) = P(F|E)
   Probability of being flagged given the application does not contain an error = (Flagged|No Error) = P(F|NE)

   b. i. P(Error & Flagged) = P(F&E)
   P(F|E) = P(F&E)/P(E)
   0.8*0.05 = P(F&E)
   P(F&E) = 0.04
   Proportion of applications that contain a material error and are flagged = 4%

   ii. P(Error & Not Flagged) = P(E &NF)
   P(NF|E) = P(NF&E)/P(E)
   P(NF&E) = P(E)*P(NF|E)
   P(NF|E) = 1 – P(F|E)
   P(NF|E) = 1-0.8 = 0.2
   P(NF&E) = 0.05*0.2
   P(NF&E) = 0.01
   Proportion of applications that contain a material error and are not flagged = 1%

   iii. P(No Error & Flagged) = P(F & NE)
   P(F|NE) = P(F&NE)/P(NE)
   0.1*0.95 = P(F&NE)
   P(F&NE) = 0.095
   Proportion of applications that do not contain a material error and are flagged = 9.5%

   iv. P(No error & Not Flagged) = P(NF &NE)
   P(NF|NE) = P(NF&NE)/P(NE)
   P(NF|NE)*P(NE) = P(NF&NE)
   P(NF|NE) = 1-P(F|NE)
   P(NF|NE) = 1-0.1
   P(NF|NE) = 0.9
   P(NF&NE) = 0.9*0.95
   P(NF&NE) = 0.855
   Proportion of applications that do not contain a material error and are not flagged = 85.5%

c. If an application is flagged, what is the probability that it contains a material error = P(E|F)
P(E|F) = P(F|E)*P(E) / [P(F|E)*P(E) + P(F|NE)*P(NE)]
P(E|F) = (0.8*0.05)/[0.8*0.05 + 0.1*0.95)
P(E|F) = 0.04/0.04+0.095
P(E|F) = 0.2962
If an application is flagged, what is the probability that it contains a material error = 29.62%

d. This statements is incorrect as only 29.7% of the flagged application contains a material error. We found the proportion in the previous problem. Hence, the statement is factually incorrect

2. P(Compliant) = 0.3
P(Report = "C" | Compliant) = 1
P(Report = "C" | Violation) = q

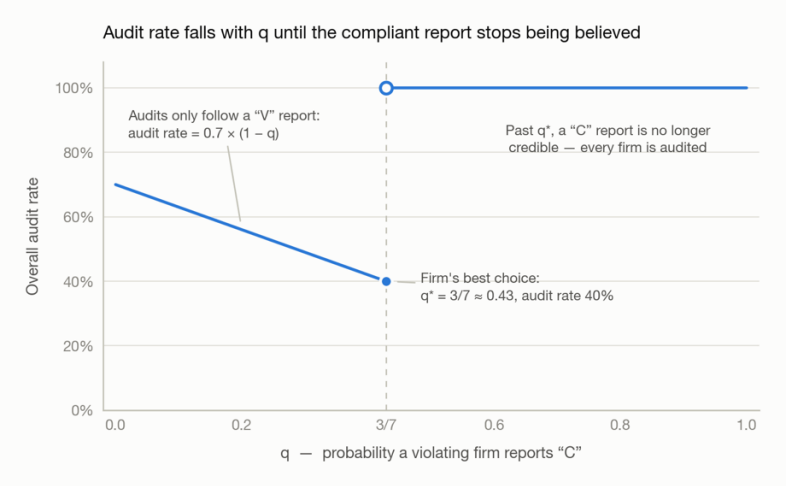
a. P(Compliant | Report = "C") = P(Report = "C"|Compliant)*P(Compliant)/[
P(Report = "C"|Compliant)*P(Compliant) + P(Report = "C"|Not Compliant)*P(Not compliant)
P(Compliant | Report = "C") = 1*0.3/(1*0.3+q*0.7)
P(Compliant | Report = "C") = 0.3/0.3+0.7q
P(Compliant | Report = "C") = P(compliant)/ P(compliant) +(1- P(compliant))q

b. If q=0.2
P(Compliant | Report = "C") = 0.3/0.3+0.14
P(Compliant | Report = "C") = 0.68
Since P(Compliant | Report = "C") > 0.5, the agency will not audit the firm that report q=0.2

c. P(Compliant | Report = "C") = 0.5
P(Compliant | Report = "C") = 0.3/0.3+0.7q
0.5 = 0.3/0.3+0.7q
0.15 + 0.35q = 0.3
q = 0.15/0.35
q = 0.4286

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d.



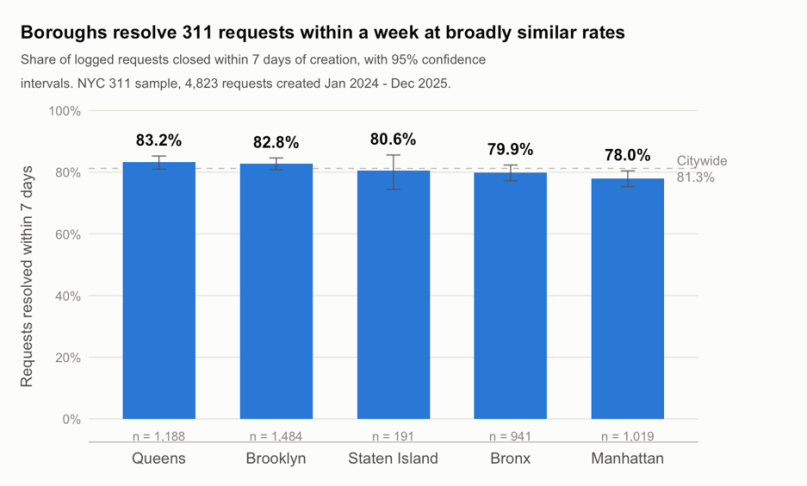
e. The audit minimising firm should chose q = 0.43. Yes, the answer is same as the one in (c). If they chose a higher q, the agency will audit the firm as the probability of firm being compliant given they reported compliant will be less than 0.5. Hence, the agency will audit the firm.

3.

```
df |>
  mutate(
    created_date = as_datetime(created_date),
    closed_date = as_datetime(closed_date),
    days_to_close = as.numeric(closed_date - created_date, units =
      "days")
  ) |>
  filter(days_to_close <= 7) |>
  group_by(borough) |>
  summarize(n = n(), .groups = "drop") |>
  mutate(prop = n / sum(n)) |>
  ggplot(aes(x = borough, y = prop)) +
  geom_col() +
  labs(
    x = "Borough",
    y = "Proportion of requests",
    title = "Proportion of 311 requests resolved within seven days, by
      borough"
  ) +
  ylim(0, 1)
```

a. This code might also use error data that might have closed date before created date. Hence, the days to close variable might give values less than 0. Hence, this is the error in the dataset. I have attached the correct graph below, which accounts for this error

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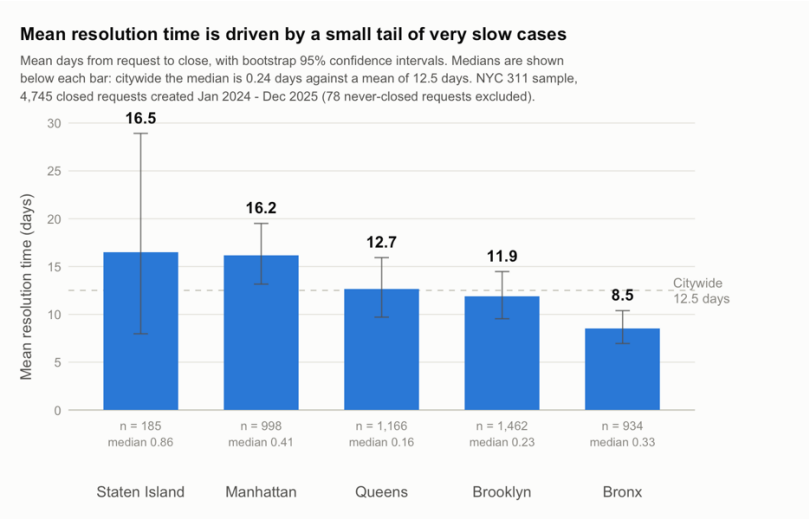


b.

```
df |>
  mutate(
    created_date = as_datetime(created_date),
    closed_date = as_datetime(closed_date),
    resolution_days = as.numeric(closed_date - created_date,
units = "days")
  ) |>
  group_by(borough, complaint_type) |>
  summarize(avg_days = mean(resolution_days, na.rm = TRUE),
.groups = "drop") |>
  group_by(borough) |>
  summarize(avg_days = mean(avg_days, na.rm = TRUE)) |>
  ggplot(aes(x = borough, y = avg_days)) +
  geom_col() +
  labs(
    x = "Borough",
    y = "Average resolution time (days)",
    title = "Average 311 resolution time by borough"
  )
```

Grouping the data by borough and complaint type changes the mean of the resolution days. As a result, we do not get the correct graph of the average resolution days. We only need to group by borough in this case. I have attached the correct graph below for your p

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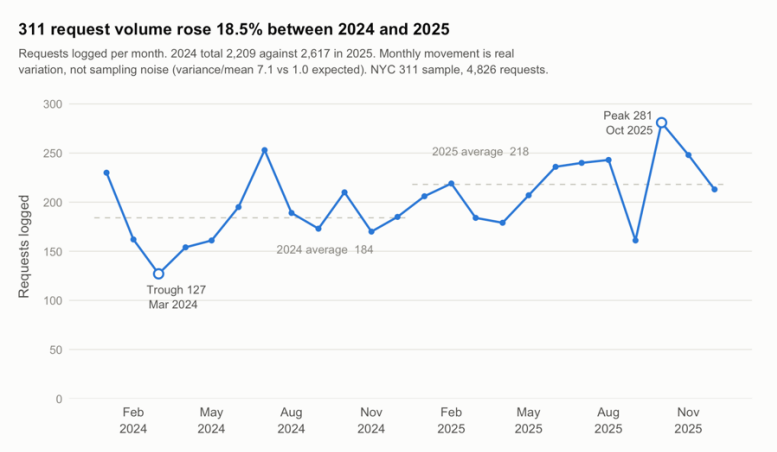


c.

```
df |>
  mutate(created_date = as_datetime(created_date)) |>
  filter(
    created_date >= ymd("2024-01-01"),
    created_date <= ymd("2025-12-31")
  ) |>
  mutate(
    month = factor(
      month(created_date, label = TRUE, abbr = TRUE),
      levels = month.abb
    )
  ) |>
  group_by(month, .drop = FALSE) |>
  summarize(n = n(), .groups = "drop") |>
  ggplot(aes(x = month, y = n, group = 1)) +
  geom_line() +
  geom_point() +
  labs(
    x = "Month",
    y = "Number of 311 requests",
    title = "311 requests by month, January 2024–December 2025"
  )
```

Mutate command is used to transform a variable which already exists. A varioable named month doesn't exist. So, we cannot mutate it to get the number of requests made by month. I have dropped the mutate command and attached the correct graph below

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4.

a. The Agent looked at the modal handling agency, descriptor values and top location type to cluster the different cluster type. The main assumption it used was who acts and under what policy authority. For instance blocked driveway (enforcement against a parked vehicle) sits with parking, while street condition (the roadway itself) sits with infrastructure.

   The agent also chose to keep noise as a separate complaint cluster despit it belonging to two different agencies. It made this choice because it considers noise as the same problem, just being responded to by different responders. It also put Oil & Gas spill as environmental hazard, for-hire vehicle complaints as Traffic, and School maintenance as building condition issue.

   In case of ambiguous complaint types, the agent used descriptor value or location type to cluster the complaint. For instance General complaint with location residential building made it into ordinary housing maintenance category

b. I changed a few categories. Firstly, I shifted Root/Sewer/Sidewalk Condition from Parks, Trees & Green Space to Public health & consumer protection, as I thought these posed risks to public health. Secondly, I moved For Hire Vehicle Complaint from Public Health & Consumer Protection to streets & public infrastructure as I felt that these were street issues. I kept rest of the categories as same

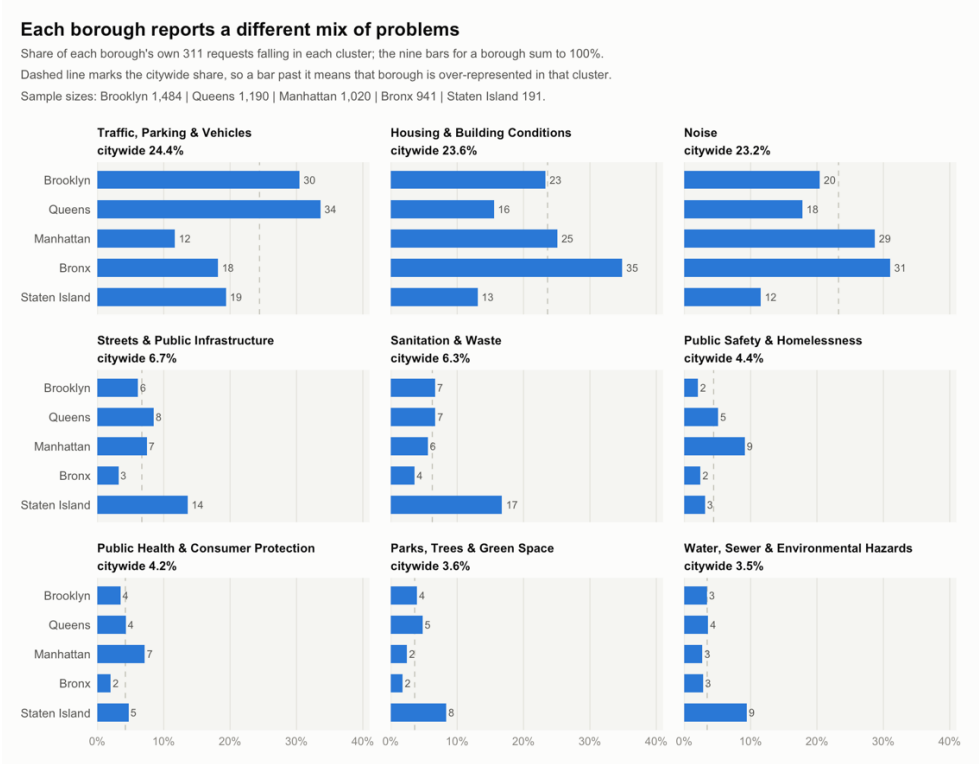
c.

Cluster	What it covers	Representative complaint types
Traffic, Parking & Vehicles	Complaints about how vehicles are parked, left or driven — enforcement aimed at a vehicle rather than at the condition of the road.	Illegal Parking; Blocked Driveway; Abandoned Vehicle

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Cluster	What it covers	Representative complaint types
Housing & Building Conditions	Habitability and maintenance failures inside residential buildings, plus construction, permit and occupancy enforcement.	HEAT/HOT WATER; UNSANITARY CONDITION; PLUMBING
Noise	Noise from any source, split in practice between police response to residential and street noise and DEP enforcement against construction and equipment.	Noise – Residential; Noise – Street/Sidewalk; Noise – Commercial
Streets & Public Infrastructure	The condition of the built streetscape — roadway, signals, signage, lighting and street furniture — together with for-hire vehicle service complaints.	Street Condition; Traffic Signal Condition; For Hire Vehicle Complaint
Sanitation & Waste	Collection service, street and sidewalk cleanliness, and illegal disposal enforcement; almost entirely Sanitation Department work.	Dirty Condition; Illegal Dumping; Missed Collection
Public Safety & Homelessness	Street-level disorder and vulnerability — the complaints that draw a police or outreach response rather than an inspection.	Encampment; Homeless Person Assistance; Drug Activity
Public Health & Consumer Protection	Pest and vector control, food safety, and regulation of businesses and street vendors.	Vendor Enforcement; Rodent; Consumer Complaint
Parks, Trees & Green Space	The urban forest and park facilities, covering street trees as well as parkland.	Damaged Tree; Maintenance or Facility; Overgrown Tree/Branches
Water, Sewer & Environmental Hazards	The water and sewer network, plus contamination and air-quality hazards.	Water System; Sewer; Air Quality

d.



Some of the important patterns from this graph are

- Traffic, parking and vehicle complaint is the highest in the city. However, the complaints are mostly concentrated in Brooklyn & Queens borough.
- Although, Bronx has less traffic, parking and vehicle complaints, it has disproportionately high complaints in the Noise category and Housing & building conditions category.
- The Staten Island borough has large number of complaints in Streets & Public Infrastructure; and Sanitation & Waste category. It gets a disproportionately high number of complaints in these categories.

Borough differences look less like differences in service quality and more like differences in built form. Density produces noise and housing complaints; car dependence produces parking complaints; low density produces infrastructure and green-space complaints.

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MEMORANDUM

To: API-232, Computational Policy Analysis

From: Shashwat

Date: September 16, 2026

Re: Homelessness in Massachusetts, 2007–2025 — a level break, not a rising trend

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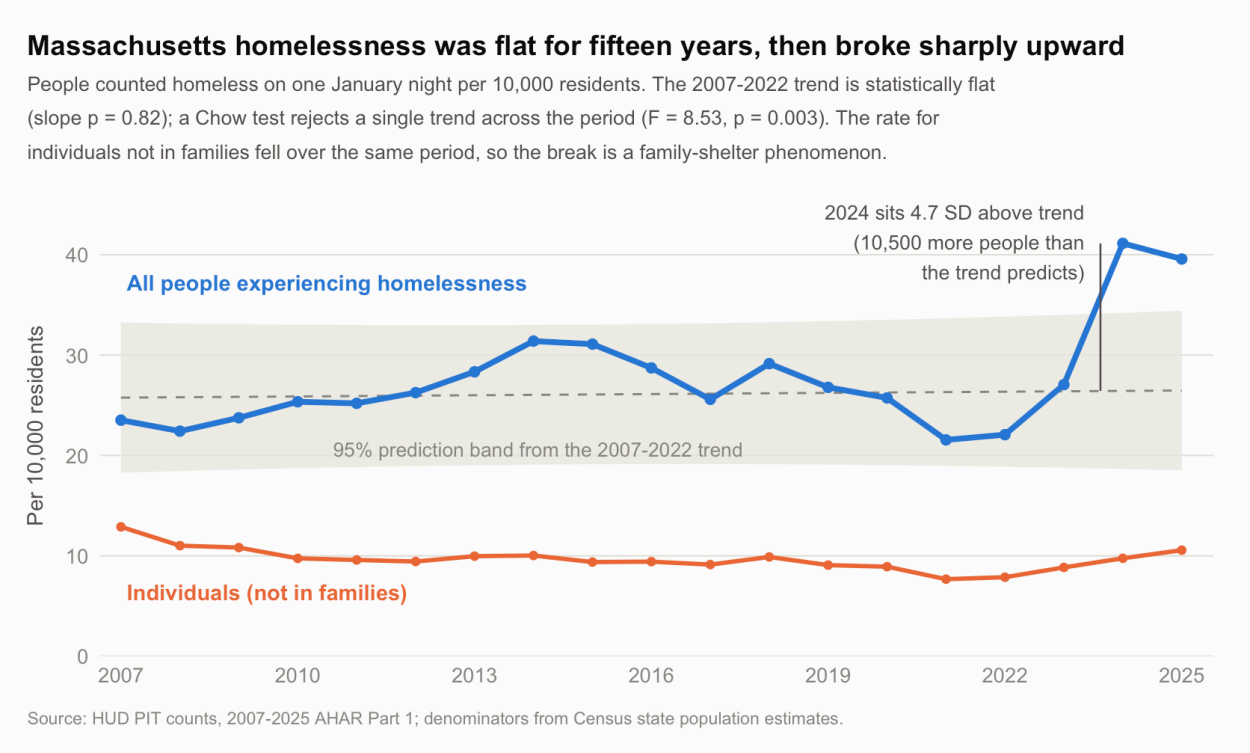
Question and why it matters

Has homelessness in Massachusetts risen steadily over the past two decades, or did something discrete happen recently? The distinction carries different policy responses. A gradual rise tracking housing costs points to supply and subsidy instruments. A sudden level shift in a state with a statutory right to shelter for families points instead to intake capacity, eligibility rules and surge financing. Massachusetts is one of only three U.S. jurisdictions guaranteeing shelter, so the question is live rather than academic.

Data and unit of observation

US Department of Housing & Urban Development (HUD) Point-in-Time (PIT) counts, published in the *2025 Annual Homelessness Assessment Report, Part 1* (huduser.gov). The **unit of observation is a state-year**: a single count of people experiencing homelessness on one night each January. That yields 19 annual observations for Massachusetts, 2007–2025, each broken down by shelter status (sheltered / unsheltered), household type (in families / individuals), chronicity and veteran status. Rates use Census Bureau state population estimates as denominators.

Analysis



The count rose from 15,127 in 2007 to 28,311 in 2025 — up 87%, or 68% per capita (23.5 to 39.6 per 10,000 residents). That headline conceals the structure. Across **2007–2022 the rate averaged 26.1 per 10,000 (Standard Deviation in 3.0, range 21.6–31.4) with no direction at all**. Almost the entire

increase arrives in two years: the count rose 89% from 2022 to 2024 (15,507 to 29,360), then eased to 28,311 in 2025. I have decomposed the increase in 13,853 homeless people below:

- **People in families:** increased by 12,418, which contributed to 89.6% of the increase
- **Sheltered:** Increased by 13,353, which amounts to 96.4% of the increase
- **Chronically homeless individuals:** increased by a meagre 426, which accounts for 3.1% total increase
- **Veterans:** increased by 16, proportionate to 0.1% increase

Two further facts sharpen this. Homelessness among **individuals not in families fell** over the full period, from 12.9 to 10.6 per 10,000, which is a statistically significant decline (−0.12 per 10,000 per year, $p = 0.006$). Massachusetts remains almost entirely sheltered as 6.2% of its 2025 count was unsheltered against 35.7% nationally, the second-lowest share among 54 states and territories, behind only New York. The interesting fact is that both states have a large right-to-shelter jurisdiction.

Testing trend against break

Because the question is precisely whether the series follows a trend or shifts level, I fit a linear trend to 2007–2022 only ($n = 16$) and tested the later years against it. This is a description of the series' structure, not a causal model, and it is reported as such.
The pre-2023 slope is +0.04 per 10,000 per year (SE 0.17; $p = 0.82$; $R^2 = 0.004$): year explains essentially none of the variation, so the earlier period is flat rather than merely slow-rising. Extrapolating that trend predicts 26.4 per 10,000 for 2024. The actual value is **41.1 — outside the interval, 4.7 residual standard deviations above the prediction, about 10,500 more people than the trend implies**. A Chow test rejects a single trend across the period ($F(2,15) = 8.53$, $p = 0.003$). Note that 2023 itself (27.1) falls inside the band: the break is 2024, not 2023.

Analysis Results Interpretation

Following things are supported by these data. Firstly, Massachusetts had no detectable upward trend in homelessness from 2007 to 2022, so the common framing that homelessness has been rising for the decade does not fit this state. Secondly, the increase is a discrete level shift in 2024, distinguishable from trend at conventional significance. Thirdly, data shows that families are overwhelmingly entering shelter, while chronic and individual homelessness were close to unaffected.

Facts not established by the data. The cause of the sudden increase in homelessness is unexplained. The timing coincides with large asylum-seeker arrivals meeting the state's right-to-shelter obligation, and the 2025 decline follows tightened Emergency Assistance eligibility. However, a single-state series of 19 points cannot attribute the break to either. Doing so would require a comparison group (other states, or the other right-to-shelter jurisdictions) and covariates for housing costs, mortgage rates, and labour markets. Separately, a right-to-shelter system *converts hidden homelessness into counted and sheltered homelessness*, so part of the measured rise may reflect enrolment in a system rather than new hardship. The HUD PIT data cannot separate the two.

Measurement limits. PIT data captures one January night and so misses episodic homelessness across the year (AHAR Part 2 annual counts would be needed). Unsheltered figures rest on street canvasses with a known undercount, so year-to-year movements in that series are the least reliable

Sources. HUD, 2025 AHAR Part 1, PIT Counts by State 2007–2025 (huduser.gov/portal/datasets/ahar). U.S. Census Bureau, State Population Totals (NST-EST2025, nst-est2020, st-est00int). Analysis in R; scripts and the extracted Massachusetts series accompany this memo.

component. Population denominators splice three Census vintages (2000–2010 intercensal, Vintage 2020, Vintage 2025), introducing small discontinuities at the 2010 and 2020 boundaries.

Conclusion (Without Claude)

My question is relevant to decision- making because homelessness has been increasing rapidly in the US over the last few years. There has been a growing policy debate regarding whether these are chronic cases or seasonal ones. Massachusetts, being a state with strong shelter jurisdiction gives an idea about the attention we need to devote towards this problem. The sudden rise in homelessness in 2024, and its continued momentum in 2025 shows that we need to further evaluate the causes behind this trend. It will allow policy makers to enact laws to break this trend.

The policy solutions can range from reigning in banks on the exorbitatant mortgage rates charged by the, controlling home prices, or interventions in the labour market to ease any economic shock for families. Hence, this question is very relevant for policy makers.
I asked the agent to extract publicly available databases on the internet that will show the homelessness rates in the state of Massachusetts. I expressed my interest in seeing homelessness rates in Massachussetts state, how it has increased over time. I also told them to explore questions around change in homelessness, variables that have caused increase in homelessness. I also told the agent that I would love to run a regression if possible.

There were no mistakes or questionable decision that I was able to identify on the part of the agent. But, I found its interpretation of the results questionable at several places. For instance I added the data that we needed to find causes of homelessness. I also asked it to avoid a regression and instead do a break test. I asked the agent to do so, because it told me that there was a clear break in homelessness rates, from 2024 onwards. I would hold on to presenting this analysis to decision makers as it currently stands. First, this memo only talks about the issue of homelessness. I would like to draft an options memo for for the policymakers, which focuses on decision options that we have. I am more inclined towards recommending them a decision path to curb homelessness. . Secondly, I don’t have data on the factors that caused

increase in homelessness. My hypotheses is that the rise in homelessness can be due to the increased cases of economic shocks in US economy after Covid-19 such as tech layoffs, rise of AI and changing labour market dynamic. I would need data on several labour market indicators to test my hypotheses. Hence, in absence of these data points, I would avoid present this analysis to policy makers.